



◆ Import

Two ways to load 'omics data into omiprep.

1 · BASE R DATA STRUCTURES

```
# build matrices yourself
samples <- data.frame('sample_id' = ...)
features <- data.frame('feature_id' = ...)
data     <- matrix(input_dat,
                    nrow = nrow(features),
                    ncol = nrow(samples),
                    dimnames = list(samples$sample_id,
                                    features$feature_id))
```

2 · COMMERCIAL FORMAT READ FUNCTIONS

```
read_metabolon(filepath, sheet, feature_sheet,
                feature_id_col, sample_sheet,
                sample_id_col, return_Omiprep)
read_nightingale(filepath, return_Omiprep)
read_olink(filepath, return_Omiprep)
read_somallogic(filepath, return_Omiprep)
```

◆ Omiprep object

Construct the S7 container for omics data, samples, and features.

```
Omiprep(data      = data,
        samples   = samples,
        features  = features)

# inspect & layer-stack helpers
summary(omiprep)
add_layer(current, layer, layer_name, force = FALSE)
batch_normalise(omiprep, run_mode_col, run_mode_colmap,
                source_layer = "input",
                dest_layer   = "batch_normalised")
```

◆ Summarise

Per-sample, per-feature, or combined summary statistics.

```
feature_summary(omiprep, source_layer,
                outlier_udist, tree_cut_height,
                feature_selection, sample_ids,
                feature_ids, features_exclude,
                output, cores, fast)

sample_summary(omiprep, source_layer,
                outlier_udist, sample_ids,
                feature_ids, output)

summarise(omiprep, source_layer, outlier_udist,
           tree_cut_height, feature_selection,
           sample_ids, feature_ids,
           features_exclude, output,
           cores, fast)
```

◆ Export

Write Omiprep data to disk as flat files, or in COMETS / MetaboAnalyst format.

```
export(omiprep, directory,
       format = "omiprep", ...)
# format: "omiprep" | "comets" | "metaboanalyst"

export_omiprep(omiprep, directory, ...)
export_comets(omiprep, directory, layer = NULL)
export_metaboanalyst(omiprep, directory,
                     layer      = NULL,
                     group_col = NULL)
```

◆ Quality control pipeline

Wrapper that runs the full omiprep QC pipeline on a layer.

```
quality_control(omiprep,
                source_layer      = "input",
                sample_missingness = 0.2,
                feature_missingness = 0.2,
                feature_skewness_threshold = NULL,
                feature_skewness_direction = "left",
                total_sum_abundance_sd = 5,
                outlier_udist = 5,
                outlier_treatment = "leave_be",
                winsorize_quantile = 1.0,
                tree_cut_height = 0.5,
                feature_selection = "max_var_exp",
                pc_outlier_sd = 5,
                max_num_pcs = 10,
                sample_ids = NULL,
                feature_ids = NULL,
                features_exclude_but_keep = NULL,
                cores = NULL,
                fast = FALSE)
```

◆ Reporting

Render an HTML or PDF QC report for an Omiprep object.

```
generate_report(omiprep, output_dir,
                output_filename = NULL,
                project = "Project",
                format = "pdf",
                template = "qc_report")
```

```
pc_and_outliers(omiprep, source_layer,
                sample_ids, feature_ids)
```

```
tree_and_independent_features(data,
                               tree_cut_height = 0.5,
                               features_exclude = NULL,
                               feature_selection = "max_var_exp",
                               cores = NULL,
                               fast = FALSE)
```

```
feature_skewness(data, threshold = NULL,
                 direction = "left")
```

```
# typical pipe
mydata |>
  quality_control(source_layer = "input") |>
  generate_report(output_dir = "out/",
                  format = "html") |>
  export("out/", format = "omiprep")
```

◆ Pipeline

High-level flow from raw vendor file to QC'd, reportable Omiprep object.

1 · Read data

- read_metabolon / nightingale
- read_olink / somalogic

2 · Pre-processing

- Normalisation
- batch_normalise()

3 · Summary statistics (raw data)

6 · Generate report & export

- generate_report()
- export()

5 · Summary statistics (filtered data)

4 · Data filtering

- Sample / feature missingness
- Feature skewness
- Total sum abundance outliers
- PCA outliers (max_num_pcs)

■ function()

Outlier treatment: leave_be · turn_NA · winsorize